```vhdl
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;
USE ieee.numeric_std.ALL;

ENTITY s16_3a IS
    PORT (
        clk : IN STD_LOGIC;
        s : IN STD_LOGIC;
        x, y : OUT STD_LOGIC
    );
END ENTITY;

ARCHITECTURE behavioral OF s16_3a IS
    SIGNAL q : STD_LOGIC_VECTOR(4 DOWNTO 0) := "00000";
    SIGNAL a, b, c, d, e, f, g, h, i, j, k, l, m, n, o : STD_LOGIC := '0';
BEGIN

    -- flipflops
    ff : PROCESS (clk)
    BEGIN
        IF rising_edge(clk) THEN
            q(4) <= c;
            q(3) <= f;
            q(2) <= i;
            q(1) <= l;
            q(0) <= o;
        END IF;
    END PROCESS;

    -- combinational logic
    a <= ((NOT s) AND q(0));
    b <= (s AND q(3));
    c <= a OR b;

    d <= (NOT s) AND q(4);
    e <= s AND q(2);
    f <= d OR e;

    g <= (NOT s) AND q(3);
    h <= s AND q(1);
    i <= g OR h;

    j <= (NOT s) AND q(2);
    k <= s AND q(0);
    l <= j OR k;

    m <= (NOT s) AND q(1);
    n <= s AND q(4);
    o <= m OR n;

    --outputs
    x <= q(1);
    y <= q(0);

END ARCHITECTURE behavioral;
```